

Human-Robot Affordance Dataset Annotation Pipeline

Section 1 Environment install

1. Install Anaconda:

<https://www.anaconda.com/download>

2. Create and activate a new conda environment:

```
conda create -n affordance python=3.10
```

```
conda activate affordance
```

3. Install required packages:

```
pip install h5py opencv-python numpy matplotlib
```

4. Install Git LFS:

```
conda install -c conda-forge git-lfs
```

```
git lfs install
```

```
HumanAndRobot — -zsh — 80x24
(9991) steven@StevendeMacBook-Pro HumanAndRobot % conda install -c conda-forge git-lfs
Channels:
 - conda-forge
 - defaults
Platform: osx-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: /opt/anaconda3/envs/9991

  added / updated specs:
    - git-lfs

The following NEW packages will be INSTALLED:

  git-lfs             conda-forge/osx-64::git-lfs-3.7.1-h990441c_1

The following packages will be UPDATED:

  ca-certificates     pkgs/main/osx-64::ca-certificates-202~ --> conda-forge/noar
```

5. Clone the Human2Robot dataset repository without downloading the full dataset:

```
GIT_LFS_SKIP_SMUDGE=1 git clone
```

<https://huggingface.co/datasets/dannyXSC/HumanAndRobot>

6. Enter the repository:

```
cd HumanAndRobot
```

7. Download only selected episodes (example):

```
git lfs pull --include="data/v0/grab_cube/episode_0.hdf5"
```

Do NOT run “git lfs pull” without “--include”, otherwise the full dataset (~233GB) may start downloading.

8. Extract paired human-robot frames:

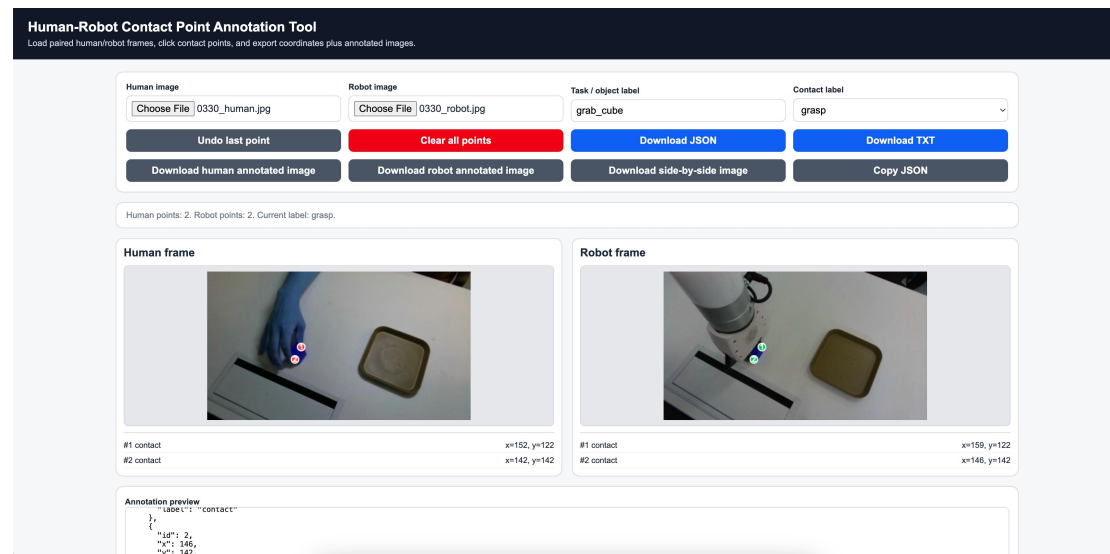
```
python extract_frames.py \
```

```
--file_path data/v0/grab_cube/episode_0.hdf5 \
```

```
--output_dir frames_ep0
```

```
HumanAndRobot — -zsh — 80x11
(9991) steven@StevendeMacBook-Pro HumanAndRobot % git lfs pull --include="data/v0/grab_cube/episode_1.hdf5"
(9991) steven@StevendeMacBook-Pro HumanAndRobot % python extract_frames.py \
--file_path data/v0/grab_cube/episode_1.hdf5 \
--output_dir frames_ep1
num frames: 366
Frames saved to: frames_ep1
```

Section 2 Annotate contact points



9. Open “contact_annotation_tool.html” in a browser.

10. Load one paired human image and robot image into the annotation tool.

11. Annotate:

- human finger/object contact point
- corresponding robot gripper/object contact point

The annotation should represent the same affordance semantics between human and robot interaction.

Example:

human grasps hammer handle ↔ robot gripper grasps hammer handle

12. Export:

- annotation JSON/TXT file
- annotated images

13. Annotation guidelines:

- use representative key frames only
- avoid blurred frames
- prefer clear contact interactions
- keep annotation consistent across tasks
- preserve original image filenames
- coordinates are stored in original image pixel coordinates